

# Engineering Reboot 2026

## Week 01 – Day 05 Field Manual

Version 2.3 (Locked Structure Compliant)

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### Mission

Complete Sprint 1 by building your first Python program and demonstrating the complete engineering workflow:

- Requirements
- Coding
- Testing
- Documentation
- Evidence Collection
- Git Version Control

Duration:

7:00 PM – 8:00 PM

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### Repository Governance Check

The following structure is already approved and shall be used without modification:

01-Week-01

- Planning
- Notes
- Journal
- Evidence
- Exercises

Day 05 exercise work shall reside in:

01-Week-01\Exercises\Day-05

No additional directories shall be created.

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# Required Deliverables

The following files must exist by end of day:

Field Manual

01-Week-01\Planning\Day-05-Field-Manual-v2.3.md

Python Program

01-Week-01\Exercises\Day-05\calculator.py

Technical Notes

01-Week-01\Notes\Day-05-Python-Calculator-Notes.md

Daily Journal

01-Week-01\Journal\Day-05-Journal.md

Evidence Log

01-Week-01\Evidence\Day-05-Evidence.md

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## End-of-Day Verification

Verify all files exist:

✓ Day-05-Field-Manual-v2.3.md

✓ calculator.py

✓ Day-05-Python-Calculator-Notes.md

✓ Day-05-Journal.md

✓ Day-05-Evidence.md

Day 05 is not complete until all files exist.

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# Task 1 – Repository Health Check

Estimated Time: 5 Minutes

Open VS Code Terminal.

Execute:

git status

Record:

- Current branch
- Working tree status

Evidence:

Screenshot of git status

Update:

Day-05-Evidence.md

Section:

Repository Health Check

Success Criteria:

Repository state understood before coding begins.

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# Task 2 – Create Day 05 Exercise Folder

Estimated Time: 2 Minutes

Verify folder exists:

01-Week-01\Exercises\Day-05

Create if absent.

No other folders are to be created.

Success Criteria:

Day-05 exercise folder exists.

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## **Task 3 – Create Calculator Program**

Estimated Time: 5 Minutes

Create:

01-Week-01\Exercises\Day-05\calculator.py

Insert:

## **Engineering Reboot 2026**

### **Week 01 Day 05**

### **Calculator Project**

Save file.

Evidence:

Screenshot of file in VS Code Explorer.

Document evidence in:

Day-05-Evidence.md

Success Criteria:

calculator.py exists.

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## **Task 4 – Implement User Input**

Estimated Time: 10 Minutes

Prompt user for:

- First Number
- Second Number

Use:

`input()`

Convert values using:

`float()`

Example:

Enter first number: 10

Enter second number: 5

Success Criteria:

Program accepts numeric input.

Evidence:

Screenshot of source code.

Update:

Day-05-Evidence.md

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## Task 5 – Implement Calculations

Estimated Time: 10 Minutes

Calculate:

- Addition
- Subtraction
- Multiplication
- Division

Store results in variables.

Example variables:

addition

subtraction

multiplication

division

Success Criteria:

All calculations execute successfully.

Evidence:

Screenshot of completed source code.

Update:

Day-05-Evidence.md

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## Task 6 – Display Results

Estimated Time: 10 Minutes

Display:

Addition

Subtraction

Multiplication

Division

Example Output:

Addition: 15

Subtraction: 5

Multiplication: 50

Division: 2

Run:

Test Case 1

10 5

Test Case 2

20 4

Test Case 3

12.5 2.5

Success Criteria:

Outputs match expected results.

Evidence:

Terminal screenshot.

Update:

Day-05-Evidence.md

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## Task 7 – Update Technical Notes

Estimated Time: 5 Minutes

Update:

01-Week-01\Notes\Day-05-Python-Calculator-Notes.md

Required Sections:

- What input() Does
- Variable Assignment
- float()
- Arithmetic Operators
- Program Flow

- Example Calculator Program

Success Criteria:

Notes completed and saved.

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## Task 8 – Complete Journal

Estimated Time: 5 Minutes

Update:

01-Week-01\Journal\Day-05-Journal.md

Required Sections:

- What I Learned
- Challenges Encountered
- Commands Used
- Lessons Learned
- Sprint 1 Reflection
- Next Steps

Success Criteria:

Journal completed.

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## Task 9 – Git Commit

Estimated Time: 5 Minutes

Execute:

git status

git add .

git commit -m "Complete Week 01 Day 05 calculator project"

git status

Success Criteria:

Working tree clean.

Evidence:

Commit screenshot.

Update:

Day-05-Evidence.md

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## Day-05-Evidence.md Minimum Sections

Repository Health Check

Calculator Creation Evidence

Input Function Evidence

Calculation Evidence

Program Execution Evidence

Git Commit Evidence

Sprint 1 Completion Summary

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## Sprint 1 Exit Criteria

✓ calculator.py completed

✓ Program executes correctly

✓ Notes completed

✓ Journal completed

✓ Evidence completed

✓ Git commit completed

✓ Repository clean

✓ Week 01 completed

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## **Sprint 1 Closure**

Day 01 – Environment Setup

Day 02 – Git Fundamentals

Day 03 – Git Workflow

Day 04 – Python Fundamentals

Day 05 – Calculator Project

Sprint 1 is complete when all deliverables have been produced, verified, committed, and stored in their approved locations.